Sentiment Analysis - A NLP Project by Prajakta Singhal

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
import re
```

```
from google.colab import files
uploaded = files.upload()

df = pd.read_csv("/content/twitter_validation.csv", encoding="ISO-8859-1", header=None)
# Assign column names
df.columns = ['Id', 'Social_media', 'target', 'text']
df
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving twitter_validation.csv to twitter_validation (4).csv
```

	Id	Social_media	target	text
0	3364	Facebook	Irrelevant	I mentioned on Facebook that I was struggling ...
1	352	Amazon	Neutral	BBC News - Amazon boss Jeff Bezos rejects clai...
2	8312	Microsoft	Negative	@Microsoft Why do I pay for WORD when it funct...
3	4371	CS-GO	Negative	CSGO matchmaking is so full of closet hacking,...
4	4433	Google	Neutral	Now the President is slapping Americans in the...
...	...	...	...	...
995	4891	GrandTheftAuto(GTA)	Irrelevant	â□ï¸□ Toronto is the arts and culture capital...
996	4359	CS-GO	Irrelevant	tHIS IS ACTUALLY A GOOD MOVE TOT BRING MORE VI...
997	2652	Borderlands	Positive	Today sucked so itâ□□s time to drink wine n pl...
998	8069	Microsoft	Positive	Bought a fraction of Microsoft today. Small wins.
999	6960	johnson&johnson	Neutral	Johnson & Johnson to stop selling talc baby po...

1000 rows × 4 columns

```
df.isna().sum()
```

```
Id              0
Social_media    0
target          0
text            0
dtype: int64
```

```
df.dtypes
```

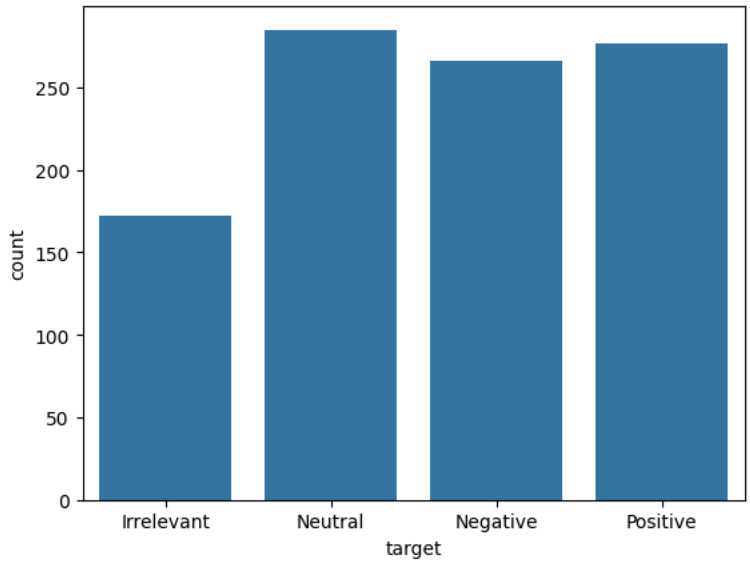
```
Id               int64
Social_media    object
target          object
text            object
dtype: object
```

```
df['target'].value_counts()
```

```
target
Neutral       285
Positive      277
Negative      266
Irrelevant    172
Name: count, dtype: int64
```

```
# Visualization
sns.countplot(x=df['target'], data=df)
```

```
<Axes: xlabel='target', ylabel='count'>
```




```
df.drop(df.index[(df['target'] == 'Irrelevant')], axis=0, inplace = True)
df
```

	Id	Social_media	target	text
1	352	Amazon	Neutral	BBC News - Amazon boss Jeff Bezos rejects clai...
2	8312	Microsoft	Negative	@Microsoft Why do I pay for WORD when it funct...
3	4371	CS-GO	Negative	CSGO matchmaking is so full of closet hacking,...
4	4433	Google	Neutral	Now the President is slapping Americans in the...
5	6273	FIFA	Negative	Hi @EAHelp Iâ□□ve had Madeleine McCann in my c...
...	...	...	...	...
993	314	Amazon	Negative	Please explain how this is possible! How can t...
994	9701	PlayStation5(PS5)	Positive	Good on Sony. As much as I want to see the new...
997	2652	Borderlands	Positive	Today sucked so itâ□□s time to drink wine n pl...
998	8069	Microsoft	Positive	Bought a fraction of Microsoft today. Small wins.
999	6960	johnson&johnson	Neutral	Johnson & Johnson to stop selling talc baby po...

828 rows × 4 columns

```
df.reset_index(drop=True, inplace=True)
df
```

	Id	Social_media	target	text
0	352	Amazon	Neutral	BBC News - Amazon boss Jeff Bezos rejects clai...
1	8312	Microsoft	Negative	@Microsoft Why do I pay for WORD when it funct...
2	4371	CS-GO	Negative	CSGO matchmaking is so full of closet hacking,...
3	4433	Google	Neutral	Now the President is slapping Americans in the...
4	6273	FIFA	Negative	Hi @EAHelp Iâ□□ve had Madeleine McCann in my c...
...	...	...	...	...
823	314	Amazon	Negative	Please explain how this is possible! How can t...
824	9701	PlayStation5(PS5)	Positive	Good on Sony. As much as I want to see the new...
825	2652	Borderlands	Positive	Today sucked so itâ□□s time to drink wine n pl...
826	8069	Microsoft	Positive	Bought a fraction of Microsoft today. Small wins.
827	6960	johnson&johnson	Neutral	Johnson & Johnson to stop selling talc baby po...

828 rows × 4 columns

```
df.drop(['Id', 'Social_media'], axis=1, inplace=True)
df
```

	target	text
0	Neutral	BBC News - Amazon boss Jeff Bezos rejects clai...
1	Negative	@Microsoft Why do I pay for WORD when it funct...
2	Negative	CSGO matchmaking is so full of closet hacking,...
3	Neutral	Now the President is slapping Americans in the...
4	Negative	Hi @EAHelp Iâ□□ve had Madeleine McCann in my c...
...	...	...
823	Negative	Please explain how this is possible! How can t...
824	Positive	Good on Sony. As much as I want to see the new...
825	Positive	Today sucked so itâ□□s time to drink wine n pl...
826	Positive	Bought a fraction of Microsoft today. Small wins.
827	Neutral	Johnson & Johnson to stop selling talc baby po...

828 rows × 2 columns

```
df['target']=df['target'].map({'Positive':1, 'Neutral':0, 'Negative': -1})
df
```

	target	text
0	0	BBC News - Amazon boss Jeff Bezos rejects clai...
1	-1	@Microsoft Why do I pay for WORD when it funct...
2	-1	CSGO matchmaking is so full of closet hacking,...
3	0	Now the President is slapping Americans in the...
4	-1	Hi @EAHelp Iâ□□ve had Madeleine McCann in my c...
...	...	...
823	-1	Please explain how this is possible! How can t...
824	1	Good on Sony. As much as I want to see the new...
825	1	Today sucked so itâ□□s time to drink wine n pl...
826	1	Bought a fraction of Microsoft today. Small wins.
827	0	Johnson & Johnson to stop selling talc baby po...

828 rows × 2 columns

```
df.dtypes
```

```
target     int64
text      object
dtype: object
```

```
tweets = df.text
tweets
```

```
0      BBC News - Amazon boss Jeff Bezos rejects clai...
1      @Microsoft Why do I pay for WORD when it funct...
2      CSGO matchmaking is so full of closet hacking,...
3      Now the President is slapping Americans in the...
4      Hi @EAHelp Iâ□□ve had Madeleine McCann in my c...
                                ...
823    Please explain how this is possible! How can t...
824    Good on Sony. As much as I want to see the new...
825    Today sucked so itâ□□s time to drink wine n pl...
826    Bought a fraction of Microsoft today. Small wins.
827    Johnson & Johnson to stop selling talc baby po...
Name: text, Length: 828, dtype: object
```

```
# NLP (Natural Language Processing)
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
```

```
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
True
```

```
# Tokenization
from nltk.tokenize import TweetTokenizer
tk = TweetTokenizer()
tweets=tweets.apply(lambda x:tk.tokenize(x)).apply(lambda x: " ".join(x))
tweets
```

```
0      BBC News - Amazon boss Jeff Bezos rejects clai...
1      @Microsoft Why do I pay for WORD when it funct...
2      CSGO matchmaking is so full of closet hacking ...
3      Now the President is slapping Americans in the...
4      Hi @EAHelp Iâ  ve had Madeleine McCann in m...
                                   ...
823    Please explain how this is possible ! How can ...
824    Good on Sony . As much as I want to see the ne...
825    Today sucked so itâ  s time to drink wine n...
826    Bought a fraction of Microsoft today . Small w...
827    Johnson & Johnson to stop selling talc baby po...
Name: text, Length: 828, dtype: object
```

```
# Remove Special Characters
tweets = tweets.str.replace('[^A-Za-z0-9]+',' ')  #$$$$
tweets
```

```
0      BBC News - Amazon boss Jeff Bezos rejects clai...
1      @Microsoft Why do I pay for WORD when it funct...
2      CSGO matchmaking is so full of closet hacking ...
3      Now the President is slapping Americans in the...
4      Hi @EAHelp Iâ  ve had Madeleine McCann in m...
                                   ...
823    Please explain how this is possible ! How can ...
824    Good on Sony . As much as I want to see the ne...
825    Today sucked so itâ  s time to drink wine n...
826    Bought a fraction of Microsoft today . Small w...
827    Johnson & Johnson to stop selling talc baby po...
Name: text, Length: 828, dtype: object
```

```
from nltk.tokenize import word_tokenize
tweets = tweets.apply(lambda x: ' '.join([w for w in word_tokenize(x) if len(w)>3]))
tweets
```

```
0      News Amazon boss Jeff Bezos rejects claims com...
1      Microsoft WORD when functions poorly SamsungUS...
2      CSGO matchmaking full closet hacking truly awf...
3      President slapping Americans face that really ...
4      EAHelp Madeleine McCann cellar past years litt...
                                   ...
823    Please explain this possible they companies ov...
824    Good Sony much want what going right much more...
825    Today sucked time drink wine play borderlands ...
826           Bought fraction Microsoft today Small wins
827    Johnson Johnson stop selling talc baby powder ...
Name: text, Length: 828, dtype: object
```

```
# Stemming
from nltk.stem import SnowballStemmer
stemmer = SnowballStemmer('english')
tweets = tweets.apply(lambda x: [stemmer.stem(i.lower()) for i in tk.tokenize(x)]).apply(lambda x: ' '.join(x))
```

```
# Stopwords Removing
from nltk.corpus import stopwords
sw = stopwords.words('english')
tweets = tweets.apply(lambda x: [i for i in tk.tokenize(x) if i not in sw]).apply(lambda x:' '.join(x))
tweets
```

```
0      news amazon boss jeff bezo reject claim compan...
1      microsoft word function poor samsungus chromebook
2           csgo matchmak full closet hack truli aw game
3      presid slap american face realli commit unlaw ...
4      eahelp madelein mccann cellar past year littl ...
                                   ...
823    pleas explain possibl compani overcharg scam a...
824             good soni much want go right much import
825    today suck time drink wine play borderland com...
826            bought fraction microsoft today small win
827    johnson johnson stop sell talc babi powder can...
Name: text, Length: 828, dtype: object
```

```
# Vectorization - TF - IDF
from sklearn.feature_extraction.text import TfidfVectorizer

vec = TfidfVectorizer()
train_data = vec.fit_transform(tweets)
print(train_data.shape)
```

```
(828, 3663)
```

```
y = df['target'].values
y
```

```
array([ 0, -1, -1,  0, -1,  1,  1,  1, -1,  1,  1, -1,  0, -1,  1,  1, -1,
        1, -1, -1,  0, -1,  0,  0, -1, -1,  1,  1, -1,  1, -1,  0,  0,  1,
        0,  1,  0,  0,  0,  1,  0, -1, -1, -1,  0,  1, -1, -1,  1,  1,  1,
        1,  1, -1, -1,  1,  1, -1,  0, -1,  0, -1,  1, -1, -1,  1,  1,  1,
        0,  0,  0,  1,  1,  0,  1,  0, -1, -1,  0,  0, -1,  1, -1, -1, -1,
        0,  1,  0, -1,  1,  1,  0,  1,  0,  1, -1,  0,  0,  0, -1,  0, -1,
        0,  0,  1,  1,  0, -1, -1,  1, -1,  0, -1,  1,  0, -1,  0,  1,  0,
        1,  1,  0,  0,  0,  0,  1,  0,  1,  1, -1,  0,  0,  0,  0, -1,  0,
        1, -1,  0, -1,  0, -1, -1, -1,  1,  1,  1,  0,  0,  1,  0,  0,  0,
        1,  0, -1, -1,  0,  1,  1,  0,  1,  1,  0,  0, -1, -1, -1, -1,  1,
        0,  0,  1,  1,  1,  1, -1,  1,  1,  0, -1, -1, -1,  1,  1, -1, -1,
        1,  1, -1,  1,  1, -1,  1,  0, -1,  0,  0,  1, -1,  1,  1,  0,  1,
       -1, -1,  1,  1,  1,  1,  0,  0,  1, -1,  0,  1,  0, -1,  0,  0, -1,
        1,  1, -1,  0,  1,  0, -1,  0, -1,  1,  1, -1, -1, -1,  1, -1,  0,
        1,  0,  0, -1,  1, -1,  1, -1,  0,  0,  1, -1,  0, -1,  1, -1,  1,
        1,  1,  1,  1,  1, -1, -1,  1, -1,  0,  0,  0,  1,  0,  1, -1,  0,
        0,  0,  0, -1,  1, -1, -1,  1,  1,  0,  0, -1, -1, -1,  0,  1,  0,
       -1,  1,  0, -1, -1, -1,  1,  0,  0, -1,  1,  1,  0,  1,  0,  0,  1,
        1, -1,  0,  1, -1,  0, -1, -1,  1,  1,  1,  0, -1,  0,  1,  0,
        1, -1, -1, -1,  1,  0,  1, -1,  0, -1,  1,  1,  1,  1,  0,  0,  0,
       -1,  1,  1,  0, -1,  1,  0, -1, -1, -1, -1, -1,  0,  0,  0,  1,  1,
       -1, -1,  0, -1,  0,  0, -1,  1, -1,  1,  1,  1,  0,  1,  0,  0, -1,
        1,  0,  0,  0,  0,  0,  0,  0,  0, -1, -1,  1,  1,  0, -1, -1,  1,
        1, -1,  1,  1,  1,  1,  1,  0, -1,  1,  0,  0,  1,  1,  1,  1,  0,
       -1, -1, -1, -1,  0,  1, -1, -1,  1,  1,  0,  0, -1, -1,  1,  0, -1,
       -1, -1,  0,  0,  1, -1, -1, -1,  0,  0,  0, -1, -1,  1, -1,  0, -1,
        0,  1, -1,  0,  1,  1, -1,  0,  0,  1, -1, -1,  0,  0, -1,  1, -1,
        0, -1, -1, -1,  1, -1,  1, -1,  1, -1, -1,  0, -1,  0, -1,  1, -1,
        0, -1, -1,  0,  0,  1, -1,  1,  0,  0,  0,  0, -1,  0,  0,  0, -1,
       -1,  0,  1,  0,  0, -1,  0,  1,  0,  0,  0,  0,  0,  1,  0,  1,  1,
        1,  0, -1,  1,  0,  0, -1,  1,  0,  0, -1,  0, -1,  0,  1, -1,  1,
       -1, -1,  0,  0,  0,  0,  1,  1,  1, -1, -1,  0,  1,  0,  0, -1,  1,
        1,  0,  1, -1, -1,  0,  1, -1,  1, -1,  0,  1,  1,  0,  0,  0,  1,
        0, -1,  0,  0, -1,  1, -1,  0,  1,  1,  1,  1,  0, -1,  0,  1,  1,
        1,  1,  1, -1,  0,  1,  0,  0, -1, -1, -1,  0,  1,  0, -1,  1,  1,
        1,  0,  1, -1,  0, -1,  0, -1,  0,  0,  1, -1,  1,  1,  0, -1,  0,
       -1, -1, -1, -1,  1,  1,  1,  1,  0, -1, -1,  1, -1, -1,  0,  0,  1,
        0, -1,  0,  1, -1,  0,  1, -1,  0,  0,  1, -1,  0, -1,  1,  1,  0,
        1,  0,  1, -1,  0,  0,  0,  1,  0,  0, -1,  1,  0, -1, -1,  0,  0,
        1, -1, -1, -1, -1,  1,  0,  0,  1,  0, -1,  1,  1, -1,  1,  1,  0,
       -1,  0,  1,  1, -1, -1, -1,  1, -1,  0, -1,  0,  0,  1,  1, -1,  0,
        1, -1, -1, -1, -1, -1, -1, -1, -1,  0, -1,  0,  0,  0,  1,  0,  0,
        0, -1,  0,  1,  0, -1, -1,  1,  0,  1,  0,  1,  0, -1,  1,  1,  1,
        1, -1, -1,  1,  0,  0,  0,  0,  0,  0, -1, -1, -1, -1,  1, -1,  0,
        1,  0, -1,  1,  1, -1,  1,  0,  0,  1, -1,  0, -1,  0,  1,  1,  0,
       -1,  1, -1, -1,  0, -1,  0, -1,  1,  0, -1, -1,  1,  1, -1,  0, -1,
        0,  0,  0,  0,  0,  0,  1,  0,  1,  1,  1, -1,  0,  1,  0,  1,  0,
        1,  0,  1,  0, -1, -1,  1,  1,  1,  1,  0, -1,  1,  1, -1, -1, -1,
        0,  1,  0,  1,  1,  0,  1, -1,  1,  1,  1,  0])
```

Start coding or generate with AI.

```
x = train_data
x
```

```
<828x3663 sparse matrix of type '<class 'numpy.float64'>'
	with 9730 stored elements in Compressed Sparse Row format>
```

```
from sklearn.model_selection import train_test_split

# Assuming x and y are defined
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.20, random_state=42)

x_train
```

```
<662x3663 sparse matrix of type '<class 'numpy.float64'>'
	with 7819 stored elements in Compressed Sparse Row format>
```

```
# Model Creation" KNN, Naive Bayes, SVC and DecisionTree
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import BernoulliNB
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
knn = KNeighborsClassifier(n_neighbors=7)
model = BernoulliNB()
model1 = SVC()
model2 = DecisionTreeClassifier()
lst = [knn, model, model1, model2]


from sklearn.metrics import confusion_matrix, accuracy_score
for i in lst:
  print(i)
  i.fit(x_train, y_train)
  y_pred = i.predict(x_test)
  print(accuracy_score(y_test, y_pred))
  print(confusion_matrix(y_test, y_pred))
```

```
KNeighborsClassifier(n_neighbors=7)
0.40963855421686746
[[21  0 34]
 [19  1 26]
 [19  0 46]]
BernoulliNB()
0.5662650602409639
[[34  4 17]
 [12 14 20]
 [14  5 46]]
SVC()
0.5301204819277109
[[23 26  6]
 [ 5 35  6]
 [ 6 29 30]]
DecisionTreeClassifier()
0.37349397590361444
[[30 14 11]
 [13 20 13]
 [34 19 12]]
```